

R08: Transforming and subsetting data with **tidyverse**

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2025-11-10

Introduction to the tidyverse



What is the tidyverse?

The **tidyverse** is a collection of R packages designed for data science. All packages share an underlying design philosophy, grammar, and data structures.

- **ggplot2** - data visualisation
- **dplyr** - data manipulation
- **tidyr** - tidy data
- **readr** - read rectangular data
- **purrr** - functional programming
- **tibble** - modern data frames
- **stringr** - string manipulation
- **forcats** - factors
- and many more ...



Tidy data¹

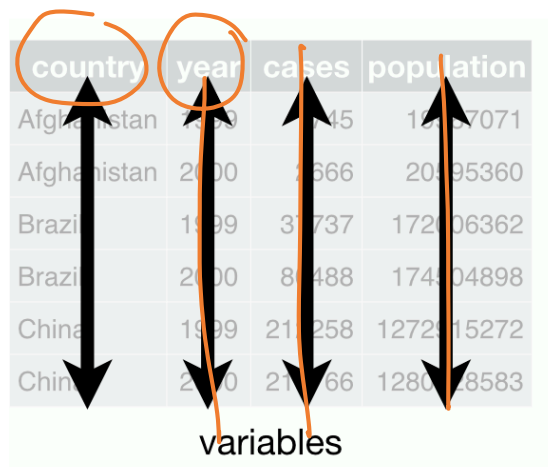


Diagram illustrating the concept of variables in tidy data. The table has four columns: country, year, cases, and population. Orange circles highlight the 'country' and 'year' headers. Vertical double-headed arrows connect the rows for each column, indicating that each column represents a variable across different observations.

country	year	cases	population
Afghanistan	1999	745	19987071
Afghanistan	2000	766	20693360
Brazil	1999	3737	17200362
Brazil	2000	8148	17460898
China	1999	21223	127291272
China	2000	21476	128423583

variables

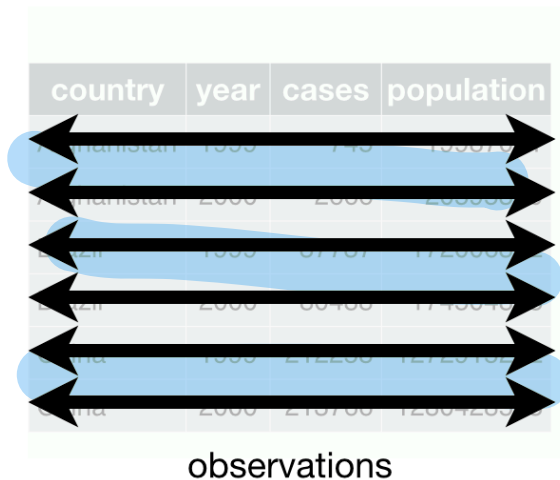


Diagram illustrating the concept of observations in tidy data. The table has four columns: country, year, cases, and population. Horizontal double-headed arrows connect the columns for each row, indicating that each row represents an observation across different variables.

country	year	cases	population
Afghanistan	1999	745	19987071
Afghanistan	2000	766	20693360
Brazil	1999	3737	17200362
Brazil	2000	8148	17460898
China	1999	21223	127291272
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observations

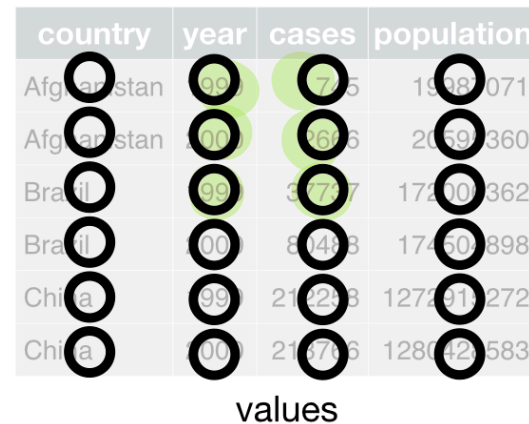


Diagram illustrating the concept of values in tidy data. The table has four columns: country, year, cases, and population. Each cell in the table is circled with a black circle, highlighting that each individual value has its own cell.

country	year	cases	population
Afghanistan	1999	745	19987071
Afghanistan	2000	766	20693360
Brazil	1999	3737	17200362
Brazil	2000	8148	17460898
China	1999	21223	127291272
China	2000	21476	128423583

values

1. Each variable must have its own column.
2. Each observation must have its own row.
3. Each value must have its own cell.

Pipe operator (**magrittr**)

- The pipe operator (`%>%`) allows us to step through sequential functions in the same way we follow if-then statements or steps from instructions

I want to find my keys, then start my car, then drive to work, then park my car.

Nested

```
1 park(drive(start_car(find("keys"))),  
2      to = "work"))
```

Piped

```
1 find("keys") %>%  
2   start_car() %>%  
3   drive(to = "work") %>%  
4   park()
```

Helpful functions for transforming and subsetting

Helpful functions for transforming and subsetting

Data transformation

- 
- `rename()`
 - `mutate()`
- 
- `pivot_longer()` and `pivot_wider()`

Data subsetting

- 
- `filter()`
 - `select()`

New dataset: `dds.discr`

- In the US, individuals with developmental disabilities typically receive services and support from state governments
 - California allocates funds to developmentally disabled residents through the Department of Developmental Services (DDS)
- Dataset `dds.discr`
 - Sample of 1,000 people who received DDS funds (out of a total of ~ 250,000)
 - Data include age, sex, race/ethnicity, and annual DDS financial support per consumer

Let's look back at the `dds.discr` dataset

- We will load the data (This is a special case! `dds.discr` is a built-in R dataset)

```
1 data("dds.discr")
```

- Now, let's take a glimpse at the dataset:

```
1 glimpse(dds.discr)
```

Rows: 1,000

Columns: 6

\$ id	<int> 10210, 10409, 10486, 10538, 10568, 10690, 10711, 10778, 1...
\$ age.cohort	<fct> 13-17, 22-50, 0-5, 18-21, 13-17, 13-17, 13-17, 13-17, 13-...
\$ age	<int> 17, 37, 3, 19, 13, 15, 13, 17, 14, 13, 13, 14, 15, 17, 20...
\$ gender	<fct> Female, Male, Male, Female, Male, Female, Female, Male, F...
\$ expenditures	<int> 2113, 41924, 1454, 6400, 4412, 4566, 3915, 3873, 5021, 28...
\$ ethnicity	<fct> White not Hispanic, White not Hispanic, Hispanic, Hispani...

Helpful functions for transforming and subsetting

Data transformation

- `rename()`
- `mutate()`
- `pivot_longer()` and `pivot_wider()`

Data subsetting

- `filter()`
- `select()`

rename (): one of the first things I usually do

- I notice that two variables have values that don't necessarily match the variable name
 - Female and male are not genders ([NIH page on sex and gender](#))
 - “White not Hispanic” combines race and ethnicity into one category ([APA page on race and ethnicity](#))

I want to rename gender to sex (not sure if assigned at birth or current sex) and rename ethnicity to R_E (race and ethnicity)

rename(): one of the first things I usually do

- `rename()` can change the name of a column
- We use: `data %>% rename(new_col_name = old_col_name)`

```
1 dds.discr1 = dds.discr %>%  
2   rename(SAB = gender,  
3          R_E = ethnicity)  
4  
5 glimpse(dds.discr1)
```

sex assigned @ birth

Rows: 1,000

Columns: 6

\$ id	<int> 10210, 10409, 10486, 10538, 10568, 10690, 10711, 10778, 1...
\$ age.cohort	<fct> 13-17, 22-50, 0-5, 18-21, 13-17, 13-17, 13-17, 13-17, 13-...
\$ age	<int> 17, 37, 3, 19, 13, 15, 13, 17, 14, 13, 13, 14, 15, 17, 20...
\$ SAB	<fct> Female, Male, Male, Female, Male, Female, Female, Male, F...
\$ expenditures	<int> 2113, 41924, 1454, 6400, 4412, 4566, 3915, 3873, 5021, 28...
\$ R_E	<fct> White not Hispanic, White not Hispanic, Hispanic, Hispani...

Helpful functions for transforming and subsetting

Data transformation

- `rename()`
- `mutate()`
- `pivot_longer()` and `pivot_wider()`

Data subsetting

- `filter()`
- `select()`

mutate(): constructing new variables from what you have

- We can create a new variable from other variables
 - Another way to say it: creates new columns that are functions of existing variables
- We often use it like:

```
1 data %>% mutate(new_variable = some_transformation_of_another_variable)
```

mutate(): create a new variable from two other variables

I want to make a variable that is the ratio of expenditures over age

```
1 dds.discr2 = dds.discr1 %>%  
2   mutate(exp_to_age = expenditures/age)  
3  
4 glimpse(dds.discr2)
```

Rows: 1,000

Columns: 7

\$ id	<int> 10210, 10409, 10486, 10538, 10568, 10690, 10711, 10778, 1...
\$ age.cohort	<fct> 13-17, 22-50, 0-5, 18-21, 13-17, 13-17, 13-17, 13-17, 13-...
\$ age	<int> 17, 37, 3, 19, 13, 15, 13, 17, 14, 13, 13, 14, 15, 17, 20...
\$ SAB	<fct> Female, Male, Male, Female, Male, Female, Female, Male, F...
\$ expenditures	<int> 2113, 41924, 1454, 6400, 4412, 4566, 3915, 3873, 5021, 28...
\$ R_E	<fct> White not Hispanic, White not Hispanic, Hispanic, Hispani...
\$ exp_to_age	<dbl> 124.2941, 1133.0811, 484.6667, 336.8421, 339.3846, 304.40...

Recoding a numeric variable into categorical

Can we recreate `age.cohort` using the `age` variable?

```
1 summary(dds.discr2)
```

id	age.cohort	age	SAB	expenditures
Min. :10210	0-5 : 82	Min. : 0.0	Female:503	Min. : 222
1st Qu.:31809	6-12 :175	1st Qu.:12.0	Male :497	1st Qu.: 2899
Median :55384	13-17:212	Median :18.0		Median : 7026
Mean :54663	18-21:199	Mean :22.8		Mean :18066
3rd Qu.:76135	22-50:226	3rd Qu.:26.0		3rd Qu.:37713
Max. :99898	51+ :106	Max. :95.0		Max. :75098

R_E	exp_to_age
White not Hispanic:401	Min. : 27.57
Hispanic :376	1st Qu.:273.88
Asian :129	Median :461.75
Black : 59	Mean : Inf
Multi Race : 26	3rd Qu.:938.12

Recoding a numeric variable into categorical (2/2)

- We can integrate other functions into `mutate()`
- For example, `case_when()` is a helpful function for mapping values to a category

Tidyverse:

```
1 dds.discr3 <- dds.discr2 %>%  
2 mutate(  
3   age.cohort2 = case_when(  
4     age <= 5 ~ "0-5",  
5     age <= 12 ~ "6-12",  
6     age <= 17 ~ "13-17",  
7     age <= 21 ~ "18-21",  
8     age <= 50 ~ "22-50",  
9     age >= 51 ~ "51+",  
10  )  
11 )
```

b/c we already
sorted ppl ≤ 5 ago,
this will only
include ppl $6 \leq \text{age} \leq 12$

when

do

Have you noticed that I change the number on `dds.discr`?

- I change the number so that R saves a new dataset
- And I do not overwrite the previous dataset
- Can be annoying, but VERY helpful when you have to go back and change code
- When you run things in real time and troubleshoot, it will be helpful to have different versions of the same dataframe

Helpful functions for transforming and subsetting

Data transformation

- `rename()`
- `mutate()`
- `pivot_longer()` and `pivot_wider()`

Data subsetting

- `filter()`
- `select()`

filter(): keep rows that match a condition

- What if I want to subset the data frame? (keep certain rows of observations)

I want to look at the data for people who between 50 and 60 years old

```
1 dds.discr4 = dds.discr3 %>%  
2   filter(age >= 50 & age <= 60)  
3  
4 glimpse(dds.discr4)
```

Rows: 23

Columns: 8

\$ id	<int> 15970, 19412, 29506, 31658, 36123, 39287, 39672, 43455, 4...
\$ age.cohort	<fct> 51+, 51+, 51+, 51+, 51+, 51+, 51+, 51+, 51+, 51+, 51+, 51...
\$ age	<int> 51, 60, 56, 60, 59, 59, 54, 57, 52, 57, 55, 52, 59, 54, 5...
\$ SAB	<fct> Female, Female, Female, Female, Male, Female, Female, Mal...
\$ expenditures	<int> 54267, 57702, 48215, 46873, 42739, 44734, 52833, 48363, 5...
\$ R_E	<fct> White not Hispanic, White not Hispanic, White not Hispani...
\$ exp_to_age	<dbl> 1064.0588, 961.7000, 860.9821, 781.2167, 724.3898, 758.20...
\$ age.cohort2	<chr> "51+", "51+", "51+", "51+", "51+", "51+", "51+", "51+", "51+", "

cannot do: $50 \leq \text{age} \leq 60$
age ≥ 50 AND
age ≤ 60
only cases that ad-
here to
both

Helpful functions for transforming and subsetting

Data transformation

- `rename()`
- `mutate()`
- `pivot_longer()` and `pivot_wider()`

Data subsetting

- `filter()` → rows

- `select()`

→ columns

`select()`: keep or drop columns using their names and types

- What if I want to remove or keep certain variables?

I want to only have age and expenditure in my data frame

```
1 dds.discr5 = dds.discr2 %>%  
2 select(age, expenditures)  
3  
4 glimpse(dds.discr5)
```

Rows: 1,000

Columns: 2

```
$ age      <int> 17, 37, 3, 19, 13, 15, 13, 17, 14, 13, 13, 14, 15, 17, 20...  
$ expenditures <int> 2113, 41924, 1454, 6400, 4412, 4566, 3915, 3873, 5021, 28...
```

Your turn

We will run a paired t-test for the cholesterol data:

1. Download `chol213_n40.csv` from the shared folder
2. Import the dataset into R
3. Create a new variable for the difference (After - Before)
4. Use `t.test()` to run a paired test

Resources

dplyr resources

- [More dplyr functions to reference!](#)

Additional details and examples are available in the vignettes:

- [column-wise operations vignette](#)
- [row-wise operations vignette](#)

and the dplyr 1.0.0 release blog posts:

- [working across columns](#)
- [working within rows](#)

R programming class at OHSU!

You can check out [Dr. Jessica Minnier's R class page](#) if you want more notes, videos, etc.

The larger tidy ecosystem

Just to name a few...

- janitor
- kableExtra
- patchwork
- gghighlight
- tidybayes

Credit to Mine Çetinkaya-Rundel

- These notes were built from Mine's notes
 - Most pages and code were left as she made them
 - I changed a few things to match our class
- Please see [her Github repository](#) for the original notes

If time

Tidy data¹

country	year	cases	population
Afghanistan	1999	745	19987071
Afghanistan	2000	2666	20693360
Brazil	1999	37737	17200362
Brazil	2000	80488	17460398
China	1999	212258	127291272
China	2000	213766	128042583

variables

country	year	cases	population
Afghanistan	1999	745	19987071
Afghanistan	2000	2666	20693360
Brazil	1999	37737	17200362
Brazil	2000	80488	17460398
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observations

country	year	cases	population
Afghanistan	99	745	19987071
Afghanistan	00	2666	20693360
Brazil	99	37737	17200362
Brazil	00	80488	17460398
China	99	212258	127291272
China	00	213766	128042583

values

1. Each variable must have its own column.
2. Each observation must have its own row.
3. Each value must have its own cell.

How do we make our data tidy??

- From a contingency table, we need to create the dataframe using the counts

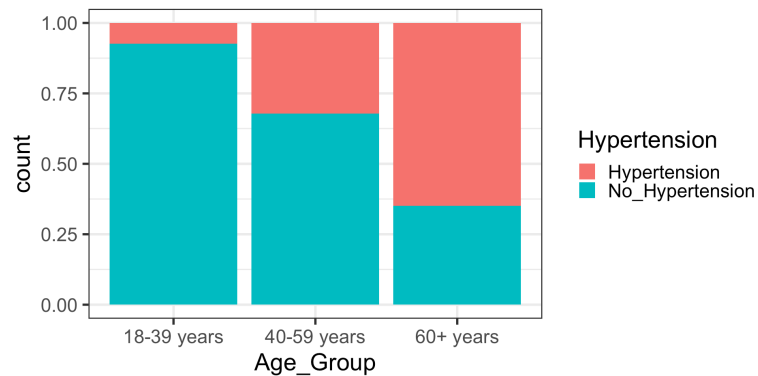
- In Lesson 4, we saw this contingency table:

Table: Contingency table showing hypertension status and age group, in thousands.

Age Group	Hypertension	No Hypertension
18-39 yrs	8836	112206
40-59 yrs	42109	88663
60+ yrs	39917	21589

- And then I magically had it in a new format so I could make this plot:

► Code





`pivot_*()` functions

wide

id	wide		
	x	y	z
1	a	c	e
2	b	d	f

I used `pivot_longer()` to create tidy data (1/2)

- Note that you won't be required to use `pivot_longer()`
 - I will give you data in a tidy form
- Here's the original data frame:

```
1 hyp_cont <- data.frame(  
2   Age_Group = c("18-39 years", "40-59 years", "60+ years"),  
3   Hypertension = c(8836, 42109, 39917),  
4   No_Hypertension = c(112206, 88663, 21589) )
```

- Note that I use `data.frame()` to make a data frame
- Then I can name each column that we saw in the contingency table
- Note that information about hypertension vs no hypertension is split between columns
 - And that we only have 3 rows of data to show all 313320 observations

I used `pivot_longer()` to create tidy data (2/2)

We need to tell `pivot_longer()`:

- Which column must be repeated (pivoted) (all other columns are not repeating)
- The name of the new column that will contain the old variable names
- Where the values in each cell under the old variables will go

```
1 hyp_data1 = pivot_longer(  
2     data = hyp_cont,  
3     cols = -Age_Group,           # columns to pivot  
4     names_to = "Hypertension",  # name of new column for variable names  
5     values_to = "Counts")       # name of new column for values  
6 hyp_data1
```

A tibble: 6 × 3

	Age_Group	Hypertension	Counts
	<chr>	<chr>	<dbl>
1	18-39 years	Hypertension	8836
2	18-39 years	No_Hypertension	112206
3	40-59 years	Hypertension	42109
4	40-59 years	No_Hypertension	88663

One more step to make it tidy

- Aka we need one more step to make it so every row is an observation
 - In this case, we want each row to represent data from one person

```
1 hyp_data = hyp_data1 %>% uncount(Counts)
2 head(hyp_data, 10)
```

```
# A tibble: 10 × 2
  Age_Group      Hypertension
  <chr>         <chr>
1 18-39 years Hypertension
2 18-39 years Hypertension
3 18-39 years Hypertension
4 18-39 years Hypertension
5 18-39 years Hypertension
6 18-39 years Hypertension
7 18-39 years Hypertension
8 18-39 years Hypertension
9 18-39 years Hypertension
10 18-39 years Hypertension
```